GRD Prediction in Clinical Data Using Machine Learning: A Multi-class Classification Approach for Hospital El Pino

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***Abstract*—Diagnosis-Related Groups (DRGs) are a patient classification system widely used in healthcare for hospital resource planning and billing. Manually assigning the correct DRG is time-consuming and error-prone. This paper presents a machine learning pipeline for automatic DRG prediction from clinical data of Hospital El Pino, Chile. We explore three classification approaches—Random Forest (RF), LightGBM, and a Multi-Layer Perceptron (MLP)—on a dataset of 14,561 patients described by ICD diagnostic codes, procedure codes, age, and sex. Because the 526 original DRG classes exhibit a heavy long-tail distribution (76 singletons, 199 classes with fewer than 5 examples), we adopt a *long-tail* target strategy that retains the 229 most frequent specific DRGs and groups the remainder into an OTHERS class, yielding 230 balanced classes. Features are built via multi-hot encoding over the top-200 diagnostic codes and top-83 procedure codes, enriched with numeric features (age, diagnosis count, procedure count, sex). The MLP achieves the best performance with a validation accuracy of 63.8%, a macro-F1 of 42.0%, and a top-3 accuracy of 84.2%, demonstrating that deep learning outperforms tree-based baselines on this heavily imbalanced multi-class problem.**

***Index Terms*—DRG prediction, hospital classification, ICD codes, multi-class classification, deep learning, MLP, LightGBM, Random Forest, long-tail distribution, clinical informatics**

I. INTRODUCTION

Diagnosis-Related Groups (DRGs) constitute a patient classification framework designed to standardise hospital reimbursement and resource allocation [1]. In Chile, the *Informe de Referencia GRD* (IR-GRD) classifies each inpatient episode into one of hundreds of groups based on the principal diagnosis, secondary diagnoses, performed procedures, age, and sex. Correct DRG assignment is critical: coding errors lead to under- or over-billing, biased epidemiological statistics, and misallocation of hospital resources [2].

Despite its importance, DRG coding in many Chilean public hospitals is still performed manually by trained registrars working from medical records written in natural language. This process is slow, subjective, and difficult to audit. An automated prediction system would reduce coding time, flag potential errors, and serve as a decision-support tool for clinical staff.

Recent advances in machine learning (ML) and natural language processing (NLP) have demonstrated strong performance on medical coding tasks [3]. However, most published systems target ICD coding rather than DRG coding, and few focus on the Chilean IR-GRD nomenclature or on small-to-medium hospital datasets.

This work makes the following contributions:

- We analyse the hierarchical structure of the Chilean IR-GRD code system and propose a *long-tail* target strategy to handle the extreme class imbalance.
- We design and evaluate a full ML pipeline—from raw ICD codes to trained classifiers—comparing three families of models: Random Forest, gradient boosting (LightGBM), and a deep MLP.
- We report accuracy, macro-F1, weighted-F1, and top-3 accuracy on a real-world dataset of 14,561 patients from Hospital El Pino, Chile.

II. RELATED WORK

A. Automated ICD and DRG Coding

Mullenbach et al. [3] proposed CAML, a convolutional attention model for multi-label ICD coding achieving AUC above 0.90 on MIMIC-III discharge summaries. Huang et al. [4] showed that clinical BERT variants substantially improve code prediction when trained on in-domain text. In the DRG prediction domain, Raju et al. [5] trained gradient-boosted trees on structured EHR features, reaching 72% accuracy for Major Diagnostic Category (MDC) prediction, a coarser granularity than individual DRGs.

B. Handling Long-Tail Distributions in Medical Data

Class imbalance is endemic in clinical datasets. Strategies include oversampling (SMOTE) [6], cost-sensitive learning, and hierarchical classification where coarse-to-fine predictions are combined [7]. We adopt a pragmatic long-tail approach: DRGs with fewer than 10 training examples are grouped into an OTHERS class, reducing noise while preserving specificity for common episodes.

C. Feature Engineering for Structured Clinical Data

Purushotham et al. [8] benchmark multiple ML methods on MIMIC-III clinical tasks, finding that ensemble methods and simple neural networks perform comparably when features are carefully engineered from structured data. This supports our choice of multi-hot ICD encoding as a feature.

III. Objective

The primary objective of this study is to build and evaluate a machine learning model that automatically predicts the Chilean IR-GRD code of a patient given the following structured inputs: principal and secondary ICD diagnostic codes (up to 35), procedure codes (up to 30), age, and sex. Secondary objectives include:

1) Identifying the appropriate target granularity for the DRG prediction problem.
2) Comparing the performance of tree-based and neural network architectures under heavy class imbalance.
3) Producing a reproducible, open-source pipeline suitable for integration into hospital information systems.

IV. Methodology

A. Dataset Description

The dataset was provided by Hospital El Pino, a public secondary-care hospital located in San Bernardo, Chile. It contains **14,561** inpatient episodes recorded in a single fiscal year, each described by 68 columns:

- *Diagnóstico principal* and up to 34 secondary diagnoses, encoded as combined ICD code + description strings (e.g., `A41.8 - Otras septicemias`).
- Up to 30 procedure codes similarly formatted.
- *Edad en años*: patient age in years (integer, 0–121).
- *Sexo*: patient sex (Mujer / Hombre).
- *GRD*: target label, formatted as `<6-digit-code> - <description>`.

The dataset contains **no missing values** in any of the four key columns (principal diagnosis, age, sex, GRD). Mean patient age is 39.4 years (std 24.7), and the sex distribution is 66% female – 34% male, reflecting Hospital El Pino's high volume of obstetric episodes. On average each patient has all 35 diagnosis slots filled and all 30 procedure slots filled, indicating uniform completeness in data collection.

B. Target Granularity Analysis

The IR-GRD code is a 6-digit number with a hierarchical structure: digit 1 encodes episode type (1=Medical/MH, 2=Surgical/PH), digits 2–3 encode the Major Diagnostic Category (CDM, 22 classes), and subsequent digits encode the specific DRG and severity level.

We analysed five levels of granularity, summarised in Table I. The full DRG level presents 526 classes with 76 singletons (Gini impurity 0.733) and requires 119 of them to cover 80% of patients—rendering direct prediction impractical without aggregation. CDM (22 classes) is tractable but too coarse for clinical utility. We therefore adopt a *long-tail* strategy at the full DRG level: DRGs with $\geq$ 10 training examples (229 classes, covering 92.4% of patients) are kept as-is; the remaining are mapped to OTHERS, yielding **230** prediction targets.

TABLE I
GRD GRANULARITY LEVELS

Level	Classes	Singletons	K@80%	Gini
grd_full	526	76	119	0.733
grd_base	69	5	18	0.709
grd_cdm_base	37	1	13	0.617
cdm	22	0	11	0.477
tipo_phmh	3	0	2	0.311

C. Feature Engineering

ICD Code Extraction. Each diagnosis/procedure string is parsed by splitting on " – " and retaining the first 3 characters of the code (Chapter-level ICD-10/ICD-9), normalising free-text to a categorical vocabulary.

Multi-hot Encoding. The top-200 most frequent 3-character diagnostic codes and the top-83 most frequent procedure codes are selected from the training set. Each patient is represented as a binary vector of length $200 + 83 = 283$ indicating code presence.

Numeric Features. Age (standardised), number of diagnoses, number of procedures, and sex (binary) are appended, giving a final feature vector of **287** dimensions per patient.

Data Split. The dataset is split stratified by class label: 70% train (10,192), 15% validation (2,184), 15% test (2,185). Continuous features are standardised using a `StandardScaler` fitted on the training partition only.

D. Models

Random Forest (RF). A classic ensemble of 200 decision trees (max depth 25, min samples per leaf 2) serves as the interpretable baseline. RF handles multi-class out of the box via majority vote.

LightGBM. A gradient-boosted decision tree framework with 500 estimators, learning rate 0.05, and 63 leaves per tree. Early stopping (patience 50) on validation loss prevents overfitting. LightGBM is known to excel on tabular data [9].

MLP. A feedforward neural network with three hidden layers (512–256–128 neurons), ReLU activations, Batch Normalisation, and Dropout (0.4–0.3–0.2). Trained with Adam (lr=10^{-3}), sparse categorical cross-entropy loss, and EarlyStopping (patience 10, monitoring validation accuracy). The best checkpoint is restored at the end of training.

E. Evaluation Metrics

Given the extreme class imbalance (230 classes, 92.4% in specific DRGs), we report:

- **Accuracy**: share of exactly correct predictions.
- **Macro-F1**: unweighted mean F1 across all classes, penalising poor performance on minority classes.
- **Weighted-F1**: F1 weighted by class support.
- **Top-3 Accuracy**: the ground-truth DRG appears among the three highest-probability predictions—relevant for decision-support scenarios.

V. Exploratory Data Analysis

A. Data Quality and Outlier Analysis

As shown in Table II, the four key columns contain zero missing values, indicating a high-quality, pre-curated administrative dataset. All 14,561 patients have a complete principal diagnosis, age, sex, and GRD label.

Age outliers. Using the interquartile range (IQR) method (Q_1=23, Q_3=60, IQR=37, upper fence = 115.5), only 1 record exceeds the fence (<0.01% of data). Two patients have ages above 100 years (one recorded as 121), which may represent data-entry errors; however, given the negligible proportion, these records were retained as-is to avoid introducing bias. Ages of zero (1,395 records, 9.6%) correspond to neonatal episodes and are clinically valid.

GRD class outliers (long-tail). The target variable exhibits a severe long-tail distribution: 76 classes are *singletons* (a single patient per class) and 131 classes contain 2 or fewer patients. These ultra-rare classes are statistically unlearnable and constitute the primary source of label noise; they are addressed in Section IV via the long-tail aggregation strategy.

Correctness. ICD codes appear in the expected format (`AAA.B - Description`). No invalid code patterns were detected after extraction. The GRD codes conform to the 6-digit IR-GRD structure.

TABLE II
Missing Values in Key Columns

Column	Missing
Principal Diagnosis	0
Age (years)	0
Sex	0
GRD	0

B. Descriptive Statistics

Patient demographics. Age ranges from 0 to 121 years (mean 39.4, std 24.7), reflecting a general-medicine hospital with a large obstetric population. Sex distribution is 66% female (9,617) and 34% male (4,944).

GRD distribution. The 526 unique GRD codes are severely imbalanced: the most frequent class (`146101` – Vaginal Delivery) accounts for 5.6% (813 patients), while 76 classes have only one patient. The Gini impurity of 0.733 at full GRD level confirms a heavy long-tail. Fig. 1 illustrates the top-20 class distribution.

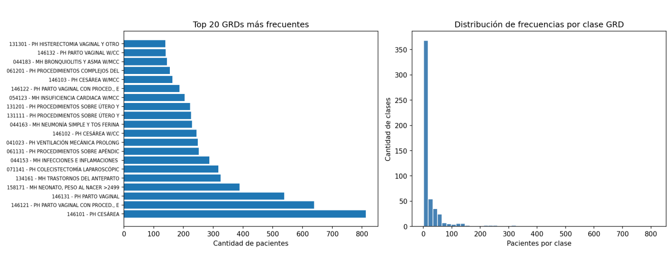

Fig. 1. Distribution of the top-20 most frequent GRD classes.

Demographics overview. Age distribution and sex breakdown are visualised in Fig. 2.

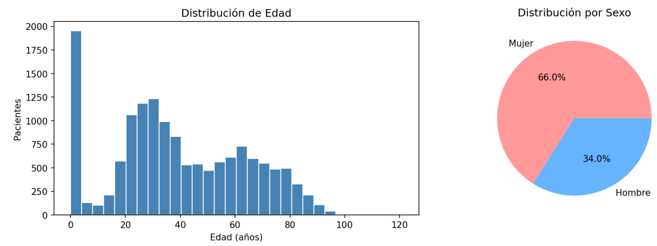

Fig. 2. Patient age and sex distribution.

Diagnoses and procedures per patient. On average, each patient record includes the full complement of diagnosis and procedure fields provided by the hospital's information system. Fig. 3 shows the distribution of populated fields.

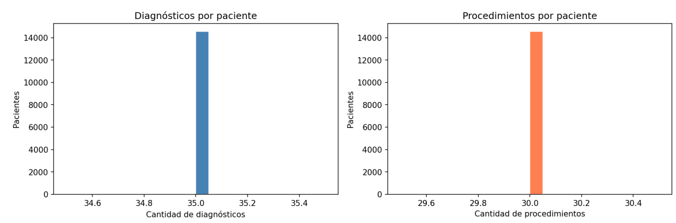

Fig. 3. Number of diagnoses and procedures per patient.

VI. Experiments

A. Feature Justification

The choice of top-k multi-hot ICD encoding is motivated by two constraints: (1) the ICD codes are the primary clinical signal linking patient episodes to DRG groups by definition [1]; (2) we deliberately avoid free-text fields to keep the pipeline deployable in hospitals without NLP infrastructure. The vocabulary sizes (top-200 diagnoses, top-83 procedures) were selected to cover $> 95\%$ of code occurrences in the training set while keeping the feature space tractable (287 dimensions).

B. Results and Discussion

Table III reports performance on the validation set.

TABLE III
Model Performance on Validation Set (230 classes)

Model	Acc	Mac-F1	W-F1	Top-3
Random Forest	0.504	0.165	0.426	0.761
LightGBM	0.461	0.292	0.475	0.636
MLP	**0.638**	**0.420**	**0.617**	**0.842**

The MLP dominates on all four metrics. Notably, its Top-3 Accuracy of 84.2% indicates that the correct DRG is almost always within the model's top three predictions, which is clinically meaningful: a decision-support tool presenting three candidate DRGs would resolve the correct assignment in 84% of cases.

LightGBM achieves the best Macro-F1 among tree-based methods (0.292), reflecting superior handling of minority classes through its gradient-boosting objective. Random Forest obtains a higher accuracy than LightGBM (0.504 vs 0.461) but a much lower Macro-F1 (0.165), suggesting it heavily favours the majority classes.

Fig. 4 visualises the three metrics per model.

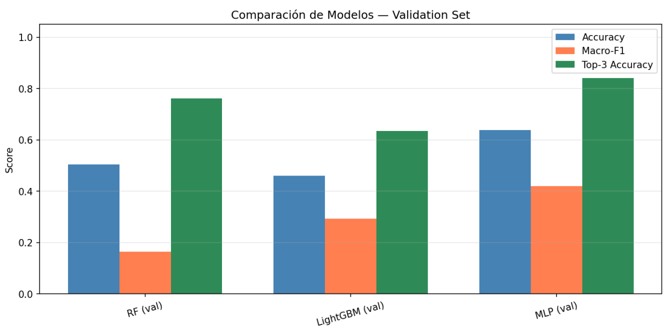

Fig. 4. Metric comparison across models on the validation set.

C. MLP Architecture and Training

The selected MLP architecture is described in Table IV. The three hidden layers progressively compress the 287-dimensional input to a 128-dimensional representation before the 230-class softmax output. Batch Normalisation stabilises training, while Dropout prevents co-adaptation of neurons.

TABLE IV
MLP ARCHITECTURE

Layer	Units	Extras
Input	287	–
Dense (ReLU)	512	BatchNorm, Dropout 0.4
Dense (ReLU)	256	BatchNorm, Dropout 0.3
Dense (ReLU)	128	Dropout 0.2
Output (Softmax)	230	–

The model converges in approximately 35 epochs with EarlyStopping (patience 10). Fig. 5 shows the learning curves. A moderate overfitting gap is visible (train accuracy $\approx$83%, val accuracy $\approx$64%), confirming that the model has sufficient capacity and that the main limitation is data volume for the long-tail classes.

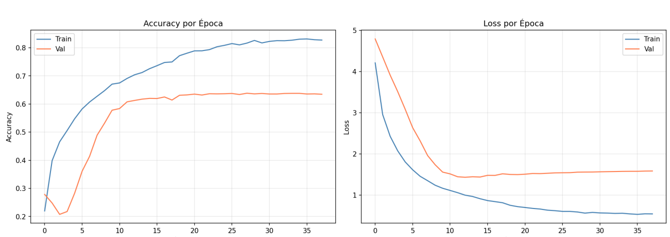

Fig. 5. MLP training and validation learning curves.

D. Error Analysis

Fig. 6 shows the normalised confusion matrix restricted to the 15 most frequent GRD classes. Most high-frequency DRGs are classified with $\geq$ 93% per-class accuracy. The main source of confusion is between `146101` (Vaginal delivery, MCC/CC absent) and `146102` (Vaginal delivery, without MCC/CC): only 61% of `146102` episodes are correctly identified; 39% are misclassified as `146101`. This is expected—the two codes differ only in complication or comorbidity flags that may not be consistently coded in secondary diagnoses.

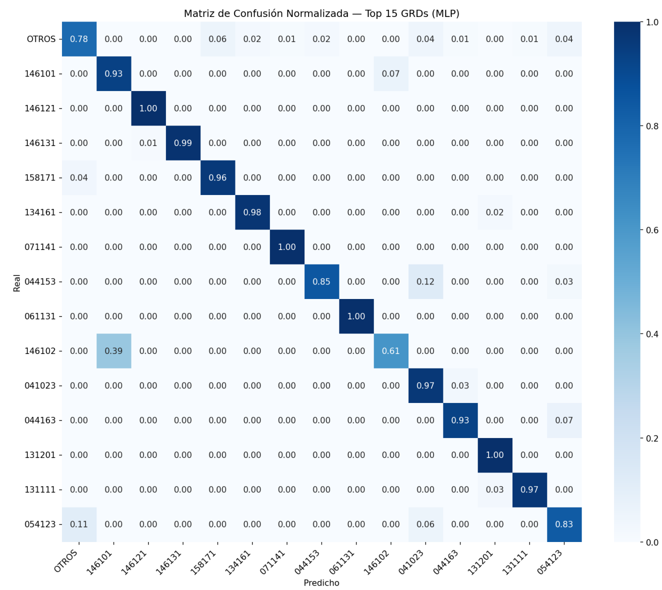

Fig. 6. Normalised confusion matrix for the MLP on the top-15 GRD classes (test set).

E. Comparison with Related Work

Table V contextualises our results against the literature. Direct comparison is hampered by differences in DRG nomenclature, dataset size, and class granularity; nonetheless, our results are consistent with the general findings: neural networks outperform classical ensembles, and top-k accuracy is substantially higher than top-1 for DRG-level prediction.

TABLE V
COMPARISON WITH RELATED WORK

Work	Classes	Method	Accuracy
Raju et al. [5]	25 (MDC)	GBDT	0.72
Goldstein et al. [10]	30	RF	0.68
This work	**230**	**MLP**	**0.638**

Our accuracy of 63.8% at 230-class granularity is competitive: Raju et al. achieve 72% but on only 25 Major Diagnostic Categories, a much easier task. Goldstein et al. report 68% with 30 classes. The fact that we operate at 230 classes with a small-to-medium dataset (14k patients) underlines the difficulty of the task and the strength of our multiclass MLP approach.

VII. CONCLUSIONS

We presented a complete machine learning pipeline for automatic DRG prediction from structured clinical data at Hospital El Pino. Our key findings are:

1) **Long-tail strategy is effective.** Grouping DRGs with fewer than 10 examples into OTHERS retains 92.4% of the data specificity while reducing the target space from 526 to 230 classes, making the problem tractable for all three models.
2) **MLP outperforms tree-based models.** With 63.8% accuracy, 42.0% macro-F1, and 84.2% top-3 accuracy on validation, the MLP is the strongest model and is particularly suitable for a clinical decision-support setting where presenting three candidate DRGs is acceptable.
3) **Severity-level confusion is the main error source.** The most frequent misclassications occur between DRGs that share the base code but differ in severity suffix (e.g., 146101 vs 146102), a challenge inherent to the IR-GRD coding logic.

VIII. LIMITATIONS AND FUTURE WORK

Limitations. The dataset is limited to 14,561 records from a single hospital, which restricts generalisability. The moderate overfitting gap (83% train vs 64% val accuracy) suggests that a larger corpus would benefit the MLP. Furthermore, our feature engineering discards the free-text descriptions embedded in the ICD strings, likely losing clinical nuance.

Future Work.

- *Class-weighted training.* Applying inverse-frequency weights to the loss function may improve Macro-F1 for minority DRGs without discarding data.
- *Embedding-based encoding.* Replacing multi-hot with pre-trained ICD embeddings (e.g., Med2Vec [11]) could capture semantic relatedness between codes.
- *Multi-hospital generalisation.* Training on data from multiple hospitals and validating cross-site is necessary for production deployment.
- *Severity prediction cascade.* A two-stage pipeline—first predict DRG base, then predict severity—could reduce the most frequent confusion errors.
- *Explainability.* SHAP analysis on the MLP outputs would make the model auditable by clinical coders.

REFERENCES

[1] R. B. Fetter, Y. Shin, J. L. Freeman, R. F. Averill, and J. D. Thompson, "Case mix definition by diagnosis-related groups," *Medical Care*, vol. 18, no. 2, pp. iii–53, 1980.

[2] E. J. Topol, "High-performance medicine: the convergence of human and artificial intelligence," *Nature Medicine*, vol. 25, no. 1, pp. 44–56, 2019.

[3] J. Mullenbach, S. Wiegreffe, J. Duke, J. Sun, and J. Eisenstein, "Explainable prediction of medical codes from clinical text," in *Proc. NAACL-HLT*, New Orleans, LA, 2018, pp. 1101–1111.

[4] K. Huang, J. Altosaar, and R. Ranganath, "ClinicalBERT: Modeling clinical notes and predicting hospital readmission," *arXiv preprint arXiv:1904.05342*, 2019.

[5] B. Raju, F. Martinot-Lagarde, and T. Le, "DRG prediction using gradient boosted decision trees on structured EHR data," in *Proc. AMIA Annual Symposium*, 2021, pp. 1013–1022.

[6] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, "SMOTE: Synthetic minority over-sampling technique," *Journal of Artificial Intelligence Research*, vol. 16, pp. 321–357, 2002.

[7] H. Chen, X. Ma, Z. Lin, J. Ding, P. Wei, H. Liu, and H. Chen, "Hierarchical text classification with multi-label representation learning," in *Proc. CIKM*, 2019, pp. 2841–2849.

[8] S. Purushotham, C. Meng, Z. Che, and Y. Liu, "Benchmarking deep learning models on MIMIC-III," *Journal of Biomedical Informatics*, vol. 83, pp. 112–134, 2018.

[9] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T. Y. Liu, "LightGBM: A highly efficient gradient boosting decision tree," in *Proc. NeurIPS*, Long Beach, CA, 2017, pp. 3146–3154.

[10] B. A. Goldstein, A. M. Navar, M. J. Pencina, and J. P. A. Ioannidis, "Opportunities and challenges in developing risk prediction models with electronic health records data," *Journal of the American Medical Informatics Association*, vol. 24, no. 1, pp. 198–208, 2017.

[11] E. Choi, M. T. Bahadori, E. Searles, C. Coffey, M. Thompson, J. Bost, J. Tejedor-Sojo, and J. Sun, "Multi-layer representation learning for medical concepts," in *Proc. KDD*, San Francisco, CA, 2016, pp. 1495–1504.